■# PAPER_246: MUGE Dual-Mode Oscillatory Gravity — Standing Wave and Hubble-Normalised Traveling Wave

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Series: Phase 2 Session 62 — §3.x Universal MUGE Sub-Term Integration

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■# PAPER_246: MUGE Dual-Mode Oscillatory Gravity — Standing Wave and Hubble-Normalised Traveling Wave

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** CondensedPhysics3.py — `MUGEDualModeOscillatoryGravityCalculator` (Session 62, grok*share*8d951e12.txt 4th-pass) **Date:** March 2026 **Series:** Phase 2 Session 62 — §3.x Universal MUGE Sub-Term Integration

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```
F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa =
5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57
```

```
g_\text{UQFF}(r) = g_\text{MUGE}(r)\cdot\Bigl(1 - [SSq]\cdot U_{b_i}\,/\,F_U(r,t)\Bigr),
\quad [SSq] = 0.57
```

Abstract

Gravity in the MUGE framework is not a static field — it supports oscillatory modes that arise from the interference of inward- and outward-propagating gravitational perturbations. This paper establishes the **dual-mode oscillatory gravity sub-term** (`g_osc`), which is the superposition of two distinct wave modes: a standing wave and a Hubble-normalised traveling wave.

Mode 1 (standing wave): `g_osc1 = 2A·cos(kx)·cos(?t)` — the classic interference pattern of two equal-amplitude counter-propagating waves. Mode 2 (traveling wave): `g_osc2 = (2p/T_H_gyr)·A·cos(kx - ?t)` — a unidirectional propagating disturbance whose amplitude is suppressed by the inverse Hubble time in gigayears, `(2p/T_H_gyr)`, connecting gravitational oscillations to the cosmological expansion rate.

The key resonance condition — `?_local = 2p/t_Hubble` — places the system at the threshold where Mode 2 dominates the superposition because `(2p/T_H_gyr) ? 1` for `T_H_gyr = 2p`. Away from resonance, Mode 1 dominates for Hubble times much larger than 2p Gyr. The time-averaged result `?g_osc? = 0` ensures no net secular drift — oscillatory gravity is a zero-mean perturbation to the static MUGE field.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?^{1}, [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. System Parameters and Equation Overview

Parameter	Symbol	Default Value	Units	Meaning
Oscillation amplitude	A_osc	1 × 10?^{1}°	m/s^{2}	Gravitational wave amplitude

Parameter	Symbol	Default Value	Units	Meaning
Wavenumber	k	1/r	1/m	Spatial frequency at system scale r
Angular frequency	?	2pc/r	rad/s	Relativistic frequency at scale r
Hubble time	t*H*gyr	13.8	Gyr	Current epoch value
Position	x	variable	m	Spatial coordinate
Time	t	variable	s	Note: in context, passed as epoch

Primary equations:

```
Mode 1 (standing wave):
g_osc1 = 2 · A · cos(k·x) · cos(?·t)
Mode 2 (Hubble-normalised traveling wave):
g_osc2 = (2p / T_H_gyr) · A · cos(k·x - ?·t)
Total:
g_osc = g_osc1 + g_osc2
Time average:
?g_osc? = 0 (both modes are zero-mean over integer wave periods)
```

Mode 2 amplitude factor at T*H*gyr = 13.8:

```
(2p / 13.8) ˜ 0.455
```

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2. Core Physics Derivation

2.1 Standing Wave — Counter-Propagating Superposition

A standing gravitational wave arises when two plane waves with equal amplitude A, wavenumber k, and frequency ? travel in opposite directions along x:

```
g? = A · cos(kx - ?t) [forward]
g? = A · cos(kx + ?t) [backward]
g_standing = g? + g? = 2A · cos(kx) · cos(?t) [trig identity]
```

This mode is spatially modulated by `cos(kx)` — nodes at `kx = (n+½)p`, antinodes at `kx = np`. At nodes, gravity is unaffected by the standing wave; at antinodes, the oscillation reaches full amplitude 2A.

2.2 Traveling Wave — Hubble-Time Amplitude Suppression

Mode 2 is a single traveling wave whose amplitude is modulated by a cosmological suppression factor:

```
g_osc2 = (2p / T_H_gyr) · A · cos(k·x - ?·t)
```

The factor `(2p / T_H_gyr)` has units of `1/Gyr` when T*H*gyr is in gigayears, but since A is already in m/s^2, the result is dimensionally consistent only if T*H*gyr is treated as a dimensionless ratio `T_H / (1 Gyr)`. This convention is standard in cosmological normalisation within MUGE.

Physical interpretation: Gravitational disturbances traveling at cosmological speeds are attenuated by the Hubble expansion. The factor `(2p/T_H_gyr)` is the instantaneous angular expansion rate in units of `(Gyr)?¹`, analogous to the Hubble parameter H0 = 1/t_Hubble but expressed in the natural angular frequency unit.

2.3 Resonance Condition

At resonance, the traveling-wave amplitude equals the standing-wave amplitude:

```
(2p / T_H_gyr) = 1 ? T_H_gyr = 2p ˜ 6.28 Gyr
```

For the current Universe (`T_H_gyr = 13.8`), Mode 2 amplitude factor ˜ 0.455 — Mode 2 carries about 45% of Mode 1 amplitude. At early cosmic times (`T_H_gyr ? 2p ˜ 6.3 Gyr`, i.e., z ˜ 0.5 in ?CDM), the two modes were equal in amplitude. At Mode 2 resonance (`T_H_gyr = 2p`), the interference pattern is maximally complex.

2.4 Zero Time Average

Both sinusoidal modes average to zero over a complete oscillation period:

```
?cos(kx)·cos(?t)?_t = cos(kx) · ?cos(?t)?_t = 0
?cos(kx - ?t)?_t = 0
? ?g_osc?_t = 0
```

This result ensures that oscillatory gravity is a **perturbative zero-mean correction** to the static MUGE field — it modulates gravity on timescale `2p/? = r/c` (light-crossing time of the system) but produces no secular drift in the total gravitational potential.

2.5 Wavenumber and Frequency at Astrophysical Scales

For a system of radius r, the natural wavenumber and frequency are `k = 1/r` and `? = 2pc/r`. This choice ties the oscillation period to the light-crossing time:

```
T_osc = 2p/? = r/c
```

For r = 1 kpc: T*osc ˜ 3.3 kyr — much shorter than stellar evolution timescales, so the oscillation averages out over physical processes. For r = 1 Mpc: T*osc ˜ 3.3 Myr — comparable to galaxy cluster merger timescales.

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3. Dual-Mode Zero-Mean Theorem

Theorem (MUGE Oscillatory Zero Mean): The dual-mode oscillatory sub-term `g_osc = g_osc1 + g_osc2` is a zero-mean bounded perturbation to the static MUGE field for all systems with finite r. The maximum instantaneous amplitude is `|g_osc|_max = A · (2 + 2p/T_H_gyr)`, reached when both modes constructively interfere at an antinode. No secular modification of total MUGE gravity results from this term in the time-averaged limit.

The **Hubble resonance condition** `T_H_gyr = 2p` is the unique epoch at which Mode 1 and Mode 2 have equal amplitude, producing the most complex gravitational interference pattern observable.

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4. Observational Predictions / Validation

- **Gravitational wave background:** The dual-mode structure predicts a specific spatial correlation pattern in the stochastic gravitational wave background — standing-wave nodes should appear as directions of suppressed GW strain in future pulsar timing arrays (IPTA, SKA).
- **Galaxy cluster mass oscillations:** At r ~ Mpc, T_osc ~ 3 Myr — oscillatory gravity contributes to ICM pressure waves seen in Chandra X-ray maps. The mode-2/mode-1 amplitude ratio (0.455 at z=0) is a direct probe of the Hubble constant at the cluster.

Early Universe enhancement: At z ˜ 0.5 (T*H*gyr ˜ 2p), the standing and traveling waves were equal — enhanced gravitational perturbations at this epoch may leave an imprint in the large-scale galaxy power spectrum at `k ˜ 0.1 h/Mpc`.

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5. References

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grok*share*8d951e12 validation session — dual-mode oscillatory gravity term implementation.

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